

PPF and the opportunity cost

ECON 101- Fall 2020

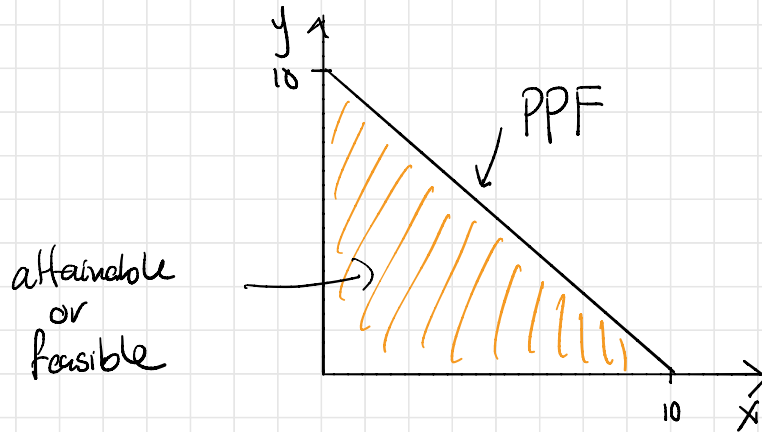


In general, the production possibilities set X

$X = \{ \text{set of all output combinations that are feasible, given current technology} \}$

Examples:

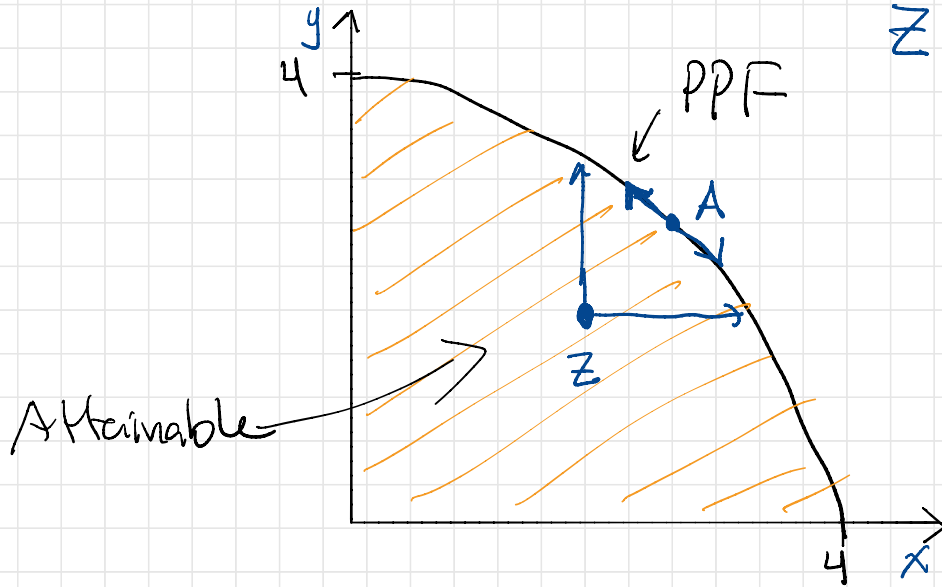
Linear Production Possibilities Set: $X + y \leq 10$



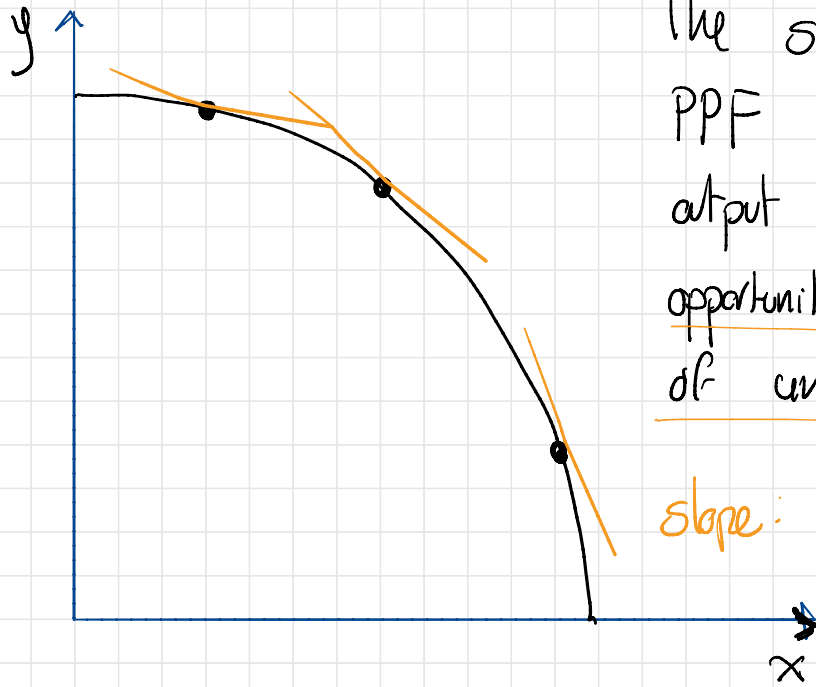
Another example:

Concave Production Possibilities Set

$$x^2 + y^2 \leq 16$$



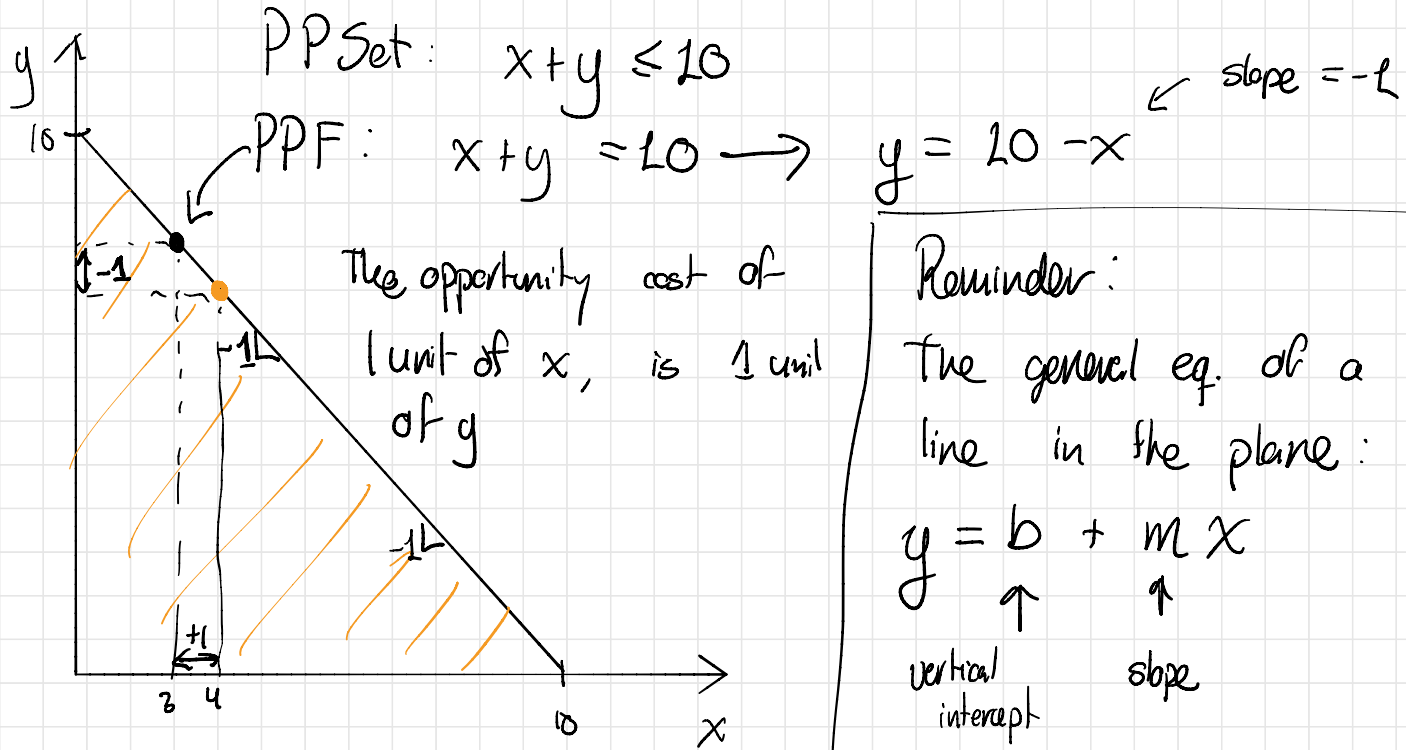
Z is not efficient!



The slope of the the PPF at a particular output combination is the opportunity cost of x in terms of units of y forgone

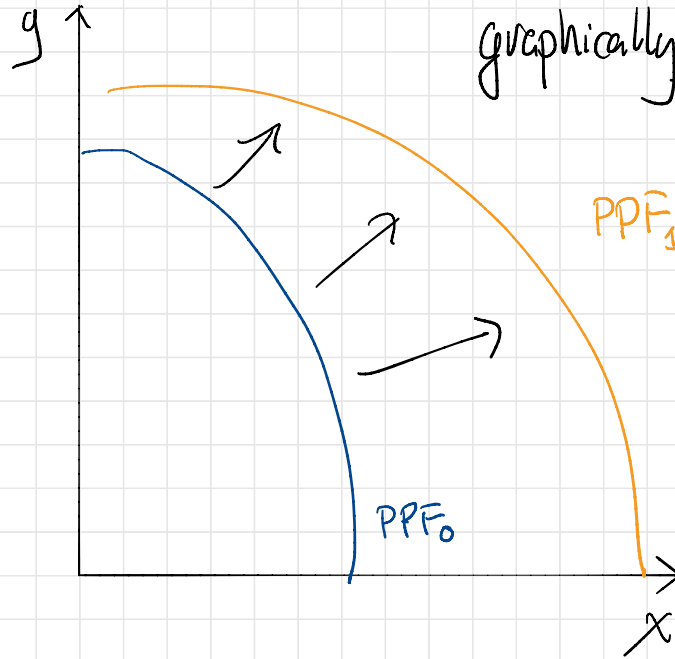
slope: OC Limit x in terms of units of y forgone

Math Reminder: the slope of a curve at a point (x, y) is the same as the slope of the tangent line at that point.

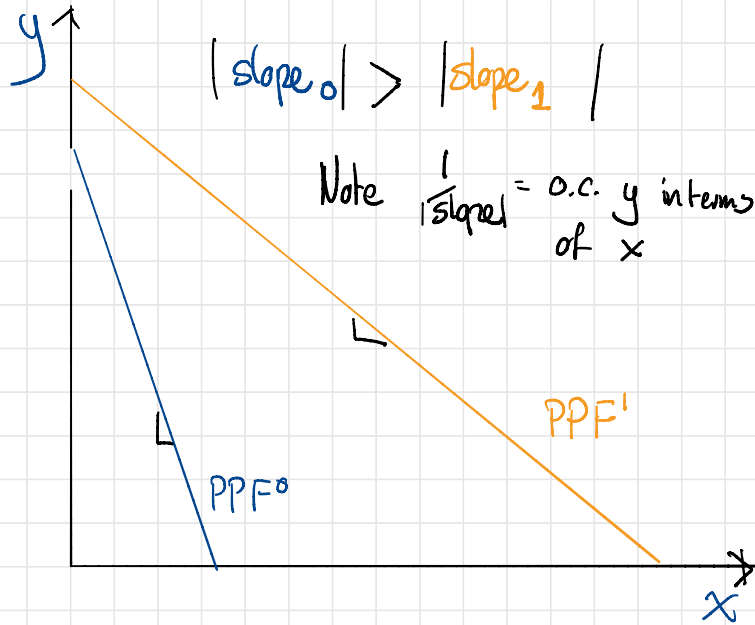


For linear PPFs, the slope is constant
 $\rightarrow$ the opportunity cost is constant

Economic Growth,



graphically, an expansion of the PPF outwards represents growth



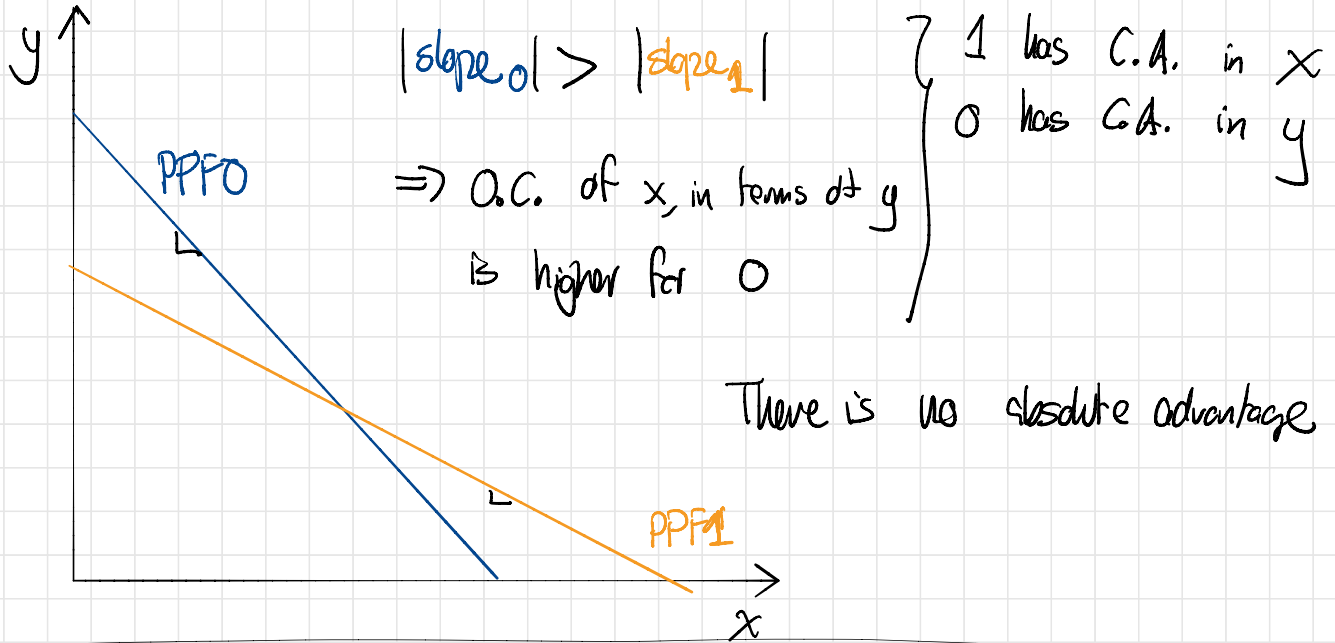
1 has a C.A. in x
 0 has a C.A. in y ,

1, using the same resources,
 is much more productive than
 0 $\Rightarrow$ 1 has an absolute
 advantage in the production
 of x and y

Comparative and
 Absolute advantage

For PPF_0 the opportunity cost of x is
 higher than for PPF_1 : 1 has a comparative
 advantage in the production of good x , 0 has C.A. in production of y

Comparative advantage but NO absolute advantage



Moral: There is always comparative advantage

Example: $x = \text{smoothies}$
 $y = \text{salads}$

60 min, Joe can produce
10 smoothies, OR 2 salads

$$60 = 10x + 2y; y = 30 - 5x$$

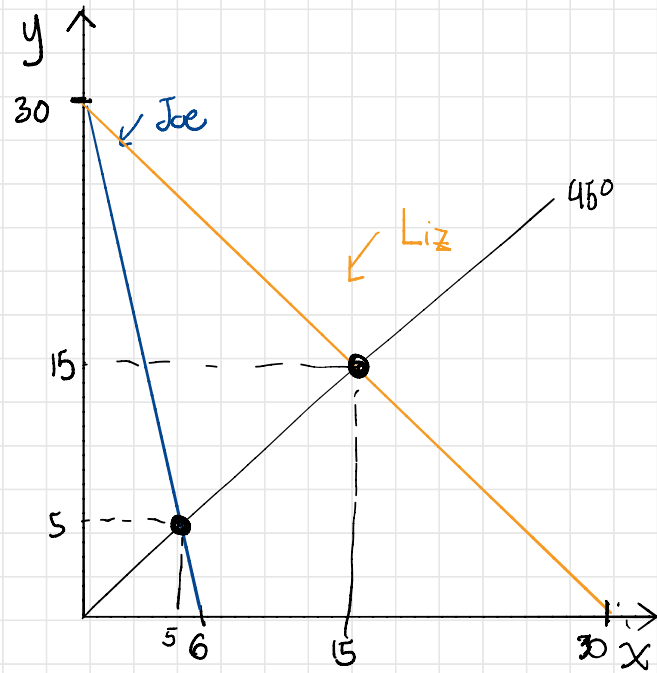
Cost of opportunity of smoothies for Joe: 5 salads

60 minutes, Liz can produce
30 smoothies / 30 salads

She takes 2 minutes to produce either of them

$$60 = 2x + 2y; y = 30 - x$$

Cost of opportunity of smoothies for Liz: 1 salad.



Liz has a comparative advantage
in smoothies; Joe has a
C.A in salad production;
Liz has an absolute advantage in
smoothie production